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21 February 1979
SUPERSEDING
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1 June 1973

MILITARY SPECIFICATION

ICE MILK, MILK SHAKE AND YOGURT MIXES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers soft serve ice milks, soft serve yogurt mixes, direct draw milk shake mixes, and dual purpose dehydrated mixes which can be reconstituted with water and frozen to provide either soft serve ice milk, yogurt or direct draw milk shakes for use by the Armed Forces as items of general issue (see 6.1).

1.2 Classification. The product shall be of the following types, classes and styles as specified (see 6.2):

Type I - Fresh soft serve ice milk mix.
Type II - Fresh direct draw milk shake mix.
Type III - Dual purpose dehydrated soft serve mix (for soft serve and direct draw milk shakes).
  Class 1 - Regular mix containing milk fat.
  Class 2 - Imitation mix, containing vegetable fat.
    Style A - Vanilla flavored.
    Style B - Chocolate flavored.
Type IV - Fresh soft serve yogurt (lowfat).
  Class 1 - Regular mix containing milk fat.
  Class 2 - Imitation mix, containing vegetable fat.
    Style A - Vanilla flavored.
    Style C - Fruit flavored (type IV only).
      1. Black cherry
      2. Black raspberry
      3. Blueberry
      4. Cherry
      5. Lemon
      6. Orange
      7. Peach
      8. Red Raspberry
      9. Strawberry

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRDNA-ES, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8910

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1163	-	Box Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Non-Resin Impregnated
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence
MIL-F-35081	-	Fat, Hydrogenated, Vegetable

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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LAWS AND REGULATIONS

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers

Bacteriological Analytical Manual

Definitions and Standards of Identity for Frozen Desserts

Definitions and Standards of Identity for Milk and Cream

Definitions and Standards of Identity for Cacao Products

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE

U.S. Standard for Grades of Butter

Dairy Plants Surveyed and Approved for USDA Grading Service

Methods of Laboratory, Analysis, Dairy Division, USDA

Grading and Inspection, General Specifications for Approved Plants and Standards for Grades of Dairy Products

(Application for copies should be addressed to the Information Division, FSQS, U.S. Department of Agriculture, Washington, DC 20250.)

Uniform Methods and Rules - Bovine Tuberculosis Eradication

Brucellosis Eradication - Recommended Uniform Methods and Rules, APHIS 91-1

United States Standards for Grades of Dry Buttermilk

United States Standards for Grades of Dry Whole Milk

United States Standards for Dry Whey

USDA Microbiological Laboratory Guide Book

(Application for copies should be addressed to Veterinary Services and Plant Health Inspection Services, US Department of Agriculture, Hyattsville, MD 20782.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

INTERNATIONAL ASSOCIATION OF MILK AND FOOD SANITARIANS, INC.

3-A Sanitary Standards

(Application for copies should be addressed to the International Association of Milk and Food Sanitarians, Inc., Box 437, Shelbyville, IL 46176.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 8th Street, N.W., Washington, DC 20036.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, DC 20044.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milks Including Methods of Analysis; Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin St., Chicago, IL 60606.)

AMERICAN OIL CHEMISTS' SOCIETY

The Journal of the American Oil Chemists' Society

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South Sixth St., Champaign, IL 61820.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for references from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material components shall comply with table I.

TABLE I. Components

Item	Requirement
1	The raw milk used in the preparation of this product shall comply with all the quality requirements for No. 1 milk specified in the Grading and Inspection, General Specifications for Approved Plants and Standards for Grades of Dairy Products. 1/ Undergrade milk shall not be used. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the US Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in: (1) A modified accredited tuberculosis area; and Either (a) A certified brucellosis-free area; or (b) Modified certified brucellosis area; or (2) An area in the process of being accredited in accordance with Uniform Methods and Rules - Bovine Tuberculosis Eradication and being certified in accordance with Brucellosis Eradication - Recommended Uniform Methods and Rules, APHIS 91-1, by the USDA. The milk shall be practically free from colostrum, fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.
2	Milk products and milk derived ingredients shall be in accordance with those permitted by the Definition and Standards of Identity for Frozen Desserts and, as applicable, shall conform to the Standards of Identity for Milk and Cream. They shall impart a pleasing and characteristic flavor to the finished product. In addition, they shall meet the requirements specified in items 3 through 8.

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TABLE I. Components (cont'd)

Item	Requirement
3	Skim milk, partially skimmed milk and cream shall be obtained by the separation of whole milk conforming to the raw milk requirements of item 1 above. Sweet cream buttermilk shall be obtained from the churning of fresh, unripened, pasteurized cream. Raw skim milk and cream shall be maintained at a temperature not higher than 45°F (7°C) and shall be used within 60 hours following separation. Pasteurized skim milk, partially skimmed milk, sweet cream buttermilk and cream shall not exceed a standard plate count of 50,000 per ml or gram (as applicable), shall be maintained at a temperature not higher than 45°F (7°C) and shall be used within 96 hours following receipt. Cream may be frozen and maintained at a temperature not higher than 0°F (-18°C).
4	Concentrated fluid dairy products or other milk derived ingredients to increase the nonfat solids content 2/ shall be free from objectionable flavors and odors. The standard plate count shall not exceed 50,000 per gram. The product shall be maintained at a temperature not higher than 45°F (7°C) and used within 96 hours following receipt.
5	Anhydrous milkfat, butteroil, plastic cream and churned fat shall meet the finished product and analytical requirements of the Grading and Inspection General Specifications for Approved Plants and Standards for Grades of Dairy Products.
6	Butter shall conform to the requirements for U.S. Grade A or better at time of use and as defined in the U.S. Standards for Grades of Butter. The butter shall be unsalted.
7	Dry milk products, such as, nonfat dry milk, dry buttermilk, dry whole milk and dry whey, shall be manufactured by the spray process and shall conform to the requirements for U.S. Extra Grade at the time of use and as defined in U.S. Standards for Grades for the product specified. The product shall have been prepared in a plant listed in USDA, "Dairy Plants Surveyed and Approved for USDA Grading Service" 1/.
8	Vegetable fat when used for class 2 product shall conform to the requirements of MIL-F-35081 except that the Wiley melting point shall be 93° ± 3°F (34° ± 2°C). Alternatively, coconut fat conforming to the requirements stated as follows may be used:

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TABLE I. Components (cont'd)

Item	Requirement
	(a) The flavor shall be bland. (b) The product shall be free from extraneous matter and the liquified product shall be clear and brilliant. (c) The color shall be 15 yellow and 1.5 red maximum (Lovibond). (d) Moisture and volatile matter shall not exceed 0.1 percent. (e) The melting point determined by the Wiley method shall be 93 ± 3°F (34° ± 2°C). (f) The peroxide value shall not exceed 1.0 meq. /kg of fat. (g) The free fatty acids shall not exceed 0.05 percent (as oleic). (h) The iodine value shall not exceed 9.0. (i) The fat stability (AOM) shall be not less than 100 hours.
9	Nutritive carbohydrate sweeteners shall be those complying with 3.6.
10	Vanilla flavoring shall be ground vanilla beans (vanilla powder), vanilla extract, imitation vanilla extract, synthetic ethyl or methyl vanillin or combinations thereof.
11	Chocolate and cocoa products used as flavoring shall meet the requirements permitted in the Definitions and Standards for Cocoa Products.
12	Fruits, preserves, fruit puree, natural fruit extracts, concentrates and other flavoring material shall be in their natural state, heat processed, dehydrated or frozen and shall be especially adapted for use in frozen yogurt. Such fruit flavors as may be used shall impart a clean, natural and pleasing flavor reaction and shall be characteristic of the applicable fruit. Imitation flavors shall not be used.
13	Colloidal stabilizers, stabilizing salts and emulsifying agents shall comply with 3.6.
14	Vitamin A palmitate, used in type III, styles A and B products, shall meet the requirements of 3.3 and shall be of the appropriate type conforming to 3.6.
15	Water shall be from a source approved as potable by State or local Health Authorities.
16	Packaging gas shall be of food grade quality and shall consist of pure nitrogen or a mixture of nitrogen and approximately 10 percent carbon dioxide plus other inert gases of the atmosphere, but shall contain no more than 0.5 percent oxygen.

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TABLE I. Components (cont'd)

Item	Requirement
17	Yogurt culture (used in type IV product) shall be a reliable, high activity, 1:1 blend of Lactobacillus bulgaricus and Streptococcus thermophilus organisms and shall be capable of producing a product meeting the finished product requirements of 3.3 (see 6.1).
18	Coloring material used shall be natural or artificial, providing it conforms to 3.6. Color shall be used only when compatible with the specific flavor and to enhance the natural color of the flavoring used if lacking in shade, hue or intensity.

1/ When Grade A milk products are required to be used for type IV product by state or local regulation such requirements shall be met (see 3.7).

2/ The use of other milk-derived ingredients is acceptable provided that the ratio of protein to total nonfat solids of the finished product (see 1.2), and the protein efficiency ratio of all protein present, shall not be decreased as a result of adding such ingredients.

3.2 Processing. Processing shall be in accordance with table II and 3.2.1.

TABLE II. Processing

Item	Requirement
1	All raw milk, unless processed within 2 hours after receipt shall be cooled immediately to 45°F (7°C) or lower and maintained at that temperature until processing is started.
2	All types of mixes shall be formulated by using the applicable materials specified in table I in the quantities normally used in recognized commercial practice provided the finished product meets the requirements of 3.3 and the exceptions specified in item 3 and 3.2.1 (also see 6.1 for recommended guidance) are met. Milk fat and vegetable fat (other than that supplied by the chocolate flavor or cocoa) shall not be used in the same mix. For type III product, all or part of the dry sweeteners and vanilla flavoring material and cocoa may be blended mechanically into the dried product. Optionally, the stabilizers or emulsifiers may be dry blended provided that this material is water-soluble and in no way affects the finished product for its intended purpose.

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TABLE II. Processing (cont'd)

Item	Requirement
3	Whey products, corn syrup solids and emulsifying agents shall not be used in type II product formulations. Buttermilk solids shall not be used in type III products.
4	The entire mix, after proper formulation (to include chocolate or cocoa flavoring, when applicable, except if dry-blended in type III product), shall be pasteurized by one of the following methods: (a) by heating every particle to at least 155°F (68°C) and holding at this temperature for at least 30 minutes; (b) by heating every particle to at least 175°F (79°C) and holding at this temperature for at least 25 seconds or (c) any other method of pasteurization recognized as equally efficient by the Food and Drug Administration, U.S. Department of Health, Education and Welfare and approved by the appropriate state or local authority. The pasteurized mix shall yield less than 1 microgram of phenol per gram when tested in accordance with the Scharer Rapid Phosphatase Method. (Note: The phosphatase test shall be considered a screening procedure and in no way shall a negative test be considered as positive proof of proper pasteurization as defined by health authority. A positive test shall be cause for immediate follow-up action to determine if improper pasteurization is directly involved). All multi-use containers, equipment and appurtenances with which dairy or other food products come into contact shall be of approved sanitary design and construction conforming to existing 3A Sanitary Standards and Accepted Practices.
5	The mix shall be homogenized effectively in accordance with good industry practice to prevent fat separation.
6	The mix shall be cooled immediately following the pasteurization/homogenization processing to a temperature of 45°F (7°C) or below and held at this temperature if the product is to be delivered within 24 hours (see item 7 for type III product and 3.4 for exception).

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TABLE II. Processing (cont'd)

Item	Requirement
7	(For type III product). During any of the processing steps prior to spray drying, if there is any delay longer than 30 minutes in the continuous flow of the fluid mix it shall be cooled to a temperature of 45°F (7°C) or below and maintained at that temperature until processing resumes. If processing delays exceed 24 hours the product shall be unacceptable. The dehydrated product shall be produced by spray drying to not less than 98 percent total solids. Spray dryers shall be of the continuous product removal type. Wet collectors may be used provided they are of sanitary design approved by local or State health authorities that they do not recycle the mix and are operated under such conditions so as to insure that the end product meets the requirements of 3.3. The appropriate vitamin A palmitate shall be blended with the dehydrated product at a level of 11,200 I.U. per pound of dry mix.

3.2.1 Processing of Type IV product. The basic dairy products and non-dairy ingredients shall be formulated, standardized and blended in accordance with good commercial practice. (See 6.1 for guidelines with respect to recommended procedure). The blend shall be subject to high temperature heat treatment (180°F to 190°F) (82°C to 88°C) for 30 minutes or its thermal equivalent, homogenized as in Table II, item 5. Following homogenization, the product shall be cooled promptly to incubation temperature (105°F to 110°F) (41°C to 43°C) or any combination of time and temperature yielding equivalent results.

3.2.1.1 Ripening and further processing. The product shall be inoculated with the yogurt culture in sufficient quantity to permit the proper development of lactic acid and the characteristic yogurt flavor (acetaldehyde) and to meet the end item requirements of 3.3 (see also 6.1). A preliminary cooling period to below 80°F (27°C) should follow before final additions of sweetener and flavors/coloring are made in order to prevent wheying-off. Complete cooling to 40°F (4°C) should follow promptly to control and prevent the development of excess acidity. In the event the product is coarse, grainy or lumpy, a texturizing procedure shall be followed to render the product smooth. The addition of fruit or preserves shall follow texturizing. If the finished product is to be shipped in frozen form, such freezing to -15° to -20°F (-26°C to -29°C) shall be accomplished within 48 hours.

3.3 Finished product. Types I, II and III (after reconstitution to a soft serve fluid mix (see 5.4.1), shall have a clean, pleasant, creamy, moderately sweet flavor, characteristic of the vanilla or chocolate flavor types used in commercial operations and classified and labeled in accordance with the Frozen Desserts Definitions and Standards of Identity (see 3.6). A mildly cooked flavor

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shall be considered acceptable. The body of the liquid mix shall be moderately viscous (see also table III item 5) and smooth, equivalent to that of fresh, heavy sweet cream. Such defects as ropy, curdled, wheyed-off watery, foamy, doughy, etc., shall be unacceptable. The texture shall be creamy and free from coarseness, lumps and graininess. Color shall be typical of the flavor (style A, B or C), uniform, pleasing and natural for fluid dairy products. Type III product (before reconstitution), shall conform to the applicable requirements of table III. Type IV product shall meet the applicable requirements for types I, II and III, and in addition shall have a clean, distinct lactic acid taste associated with the custard like yogurt of commerce. The mix shall have a pleasing nutty-like (acetaldehyde) sweet taste, but shall not mask the basic acid character of the product or vice versa. Style C product shall be of the flavor specified with sufficient flavor intensity so as to be readily and accurately identified. There shall be no off-flavors such as, but not limited to bitter, yeasty or diacetyl. The product shall have a smooth body and shall be creamy with a light to moderate viscosity similar to that of good quality commercial ice milk mix.

TABLE III. Finished product

Item	Requirement
1	Style A product shall have a white or light cream color. Style B product shall have a light to medium shade chocolate brown color.
2	The finished dehydrated product shall possess a fresh, sweet, pleasing flavor and aroma characteristic of the applicable style and after reconstitution the product shall be free from objectionable flavors and odors such as oxidized, stale, rancid, tallowy, cooked, scorched or fishy.
3	The product shall be free from dirt, insect, insect parts, rodent hairs and filth, wood, glass, paper or other foreign material and shall not be adulterated by addition of preservatives.
4	The product shall be free from lumps that do not fall apart under light pressure and shall be readily reconstituted in water at a temperature of 40° to 60°F (4° to 16°C) when agitated with a wire whisk or by other mechanical means (see 5.4.1).
5	The viscosity of the prepared and reconstituted mixes shall be such as to be readily flowable from the mix hopper of the soft serve or milk shake type freezer into the freezing cylinder.

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TABLE IV. Analytical requirements

Item	Requirement	Type I Style A	Type I Style B	Type II 1/ Style A	Type II 1/ Style B	Type III Style A	Type III Style B	Type IV Style A	Type IV Style C
1	Milk or vegetable fat (as applicable), not less than, (percent)	3.25	2.75 2/	3.25	2.75 2/	12.40	9.30 3/	1.50 4/	1.50 4/
2	Milk solids not fat, not less than, (percent) 5/	12.50	12.00	13.00	12.00	38.50	32.00	12.00	12.00
3	Moisture, not more than, (percent)	-	-	-	-	2.00	2.00	-	-
4	Standard plate count (per gram), maximum	50,000	50,000	50,000	50,000	50,000	50,000	-	-
5	Coliform, count (per gram), maximum	10	20	10	20	10	20	10	10
6	Salmonellae	-	-	-	-	Negative 6/	Negative 6/	-	-
7	Yeast and mold count/gram (maximum)	-	-	-	-	-	-	10	10
8	pH range (± .1 unit)	-	-	-	-	-	-	4.0 to 4.4	4.0 to 4.3
9	Scorched particles, not more than, (milligrams) per 25 gram sample	-	-	-	-	7.5	-	-	-

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TABLE IV. Analytical requirements (cont'd)

Item	Requirement	Type I Style A	Type I Style B	Type II Style A	Type II 1/ Style B	Type III Style A	Type III Style B	Type IV Style A	Type IV Style C
10	Oxygen in head-space gas, not more than, (percent) 7/	-	-	-	-	3.0	3.0	-	-
11	Vitamin A palmitate, I.U./lb of dry mix	-	-	-	-	11,200 I.U.	11,200 I.U.	-	-

1/ The minimum total fat content required by the state law where the product is to be consumed will apply. Where the state does not specify a minimum fat content. the requirements of this specification will apply.

2/ Exclusive of fat from chocolate or cocoa flavoring. Total fat shall have a minimum test of 3.25%. Plant records shall be maintained and made available upon request as an aid in determining milk or vegetable fat compliance.

3/ Exclusive of fat from chocolate or cocoa flavoring. Total fat shall have a minimum test of 10.70%. Plant records shall be maintained and made available upon request as an aid in determining milk or vegetable fat compliance.

4/ The flavor dilution factor shall be considered in the formulation so that the finished product meets the minimum test requirements.

5/ Plant records shall be reviewed to determine compliance.

6/ Negative in a 25 gram sample.

7/ Maximum value 7 days after packaging.

3.3.1 Frozen product requirements. Requirements for the frozen products shall be as specified in 3.3.1.1 through 3.3.1.3.

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3.3.1.1 Types I, III (after reconstitution - see 5.4.1), style A, soft serve ice milk and IV. The liquid mix when frozen in a soft serve type freezer to a drawing temperature of 18° to 22°F (-8° to -6°C), shall result in a soft serve product complying with the following characteristics:

(1) No difficulty in obtaining an overrun of 30 to 40 percent for types I and III, and 40 to 60 percent for type IV which may be calculated by the following formula:

$$\frac{A-B}{B} \times 100 = \text{percent overrun}$$

Where, A = weight of the mix (for convenience, use a straight sided, flat rimmed cup of approximately 8 oz. capacity and fill to the brim).
B = weight of same volume of frozen product.

(2) The color shall be white to light cream and have a dry, non-glistening surface appearance (type I, type III, style A and type IV, style A).

(3) The product as it emerges from the freezer shall have good peaking (coiling up) characteristics.

(4) The body shall be stiff and firm and have good resistance to melting.

(5) The texture shall be smooth and free from palpable ice crystals and the product shall show no tendency toward churning in the freezer cylinder.

(6) The flavor shall be sweet, clean and characteristically that of vanilla (type I, style A and type III, style A).

(7) Type IV, style A shall have a distinct lactic acid taste with a characteristic vanilla flavor.

(8) For type IV, style C. The flavor shall reflect the flavor character described in 3.3. Style C product may or may not show pieces of fruit in the frozen form, as applicable.

3.3.1.2 Types I and III (after reconstitution - see 5.4.1), style B, soft serve ice milk. The liquid mix when frozen in a soft serve type freezer to a drawing temperature of 18° to 22°F (-8° to -6°C), shall result in a soft serve product which complies with the same requirements specified in 3.3.1.1, except for the following differences:

(1) The color shall be light to medium chocolate brown and the flavor characteristic of a clean and distinctive chocolate flavor, without interfering flavor-carrying enhancers, such as honey, malt syrup or molasses. In addition, the chocolate shall not have an objectionable harsh or bitter background.

(2) The frozen product shall be free of "flecking", a defect which may be described as a noticeable agglomeration of cocoa fat particles.

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3.3.1.3 Types II and III (after reconstitution - see 5.4.1), direct draw milk shakes. The liquid mix when frozen in a soft serve type freezer, adapted for making milk shakes or in ready-to-serve shake make freezers to a drawing temperature of 27° to 30°F (-3° to -1°C), shall result in a milk shake complying with 3.1.1.1 (2) and (6) style A) 3.3.1.2 (1) and (2) for style A and with the following characteristics:

(1) No difficulty in obtaining an overrun of 20-30 percent, calculated in accordance with 3.3.1.1.

(2) The resulting shake shall be moderately smooth, foamy, and thick. Sufficient flexibility in the original formulation shall be provided so that, when preferred, the resultant shake can be frozen (by controlling drawing temperature and overrun) to a thicker, heavier consistency or to a light, thin product.

(3) There shall be no evidence of component separation, gumminess, stickiness, graininess or butteriness.

(4) The product shall have a refreshing quality and shall not leave a thick slimy coating in the mouth.

(5) Additionally, type III product shall possess the following physical characteristics when the reconstituted mix is blended with the addition of water in the ratios suggested:

Texture condition desired	Ratio by volume of reconstituted mix to water
Smooth and free from palpable ice crystals	no dilution
Slightly coarse	3 to 1
Medium coarse	3 to 2
Coarse with small palpable ice crystals	3 to 3

3.4 Age of product. Types I, II and IV products shall be delivered within 96 hours after date of manufacture provided product is held at 40°F (4.44°C) or lower. If type IV is in a frozen state, product shall be delivered within 10 days after date of manufacture. Unless otherwise specified by the procuring agency (see 6.2), type III product shall be shipped within 60 days subsequent to production.

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3.5 Delivery temperature of types I, II and IV products. The temperature of the product(s) at the time of delivery shall not exceed 50°F (10°C). In the case of type IV product in frozen form, the temperature of the product at any point shall not exceed -10°F (-23°C).

3.6 All deliveries shall conform in every respect to the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.7 Plant qualification. The product shall originate, be processed, handled, packaged and stored in plants meeting the requirements of MIL-STD-668 or in plants which operate under the Grading and Inspection, General Specifications for Approved Plants and Standards for Grades of Dairy Products and approved for the finished product described in this specification. When Grade A milk products are required for type IV product manufacture, the plant shall have received prior sanitary approval by the Military Veterinary Services in accordance with the Grade "A" Pasteurized Milk Ordinance or shall have a pasteurization plant compliance rating of 90 or more as certified by a State Milk Sanitation rating officer and listed in the document titled, Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers, published quarterly by the US Public Health Service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification. The product shall be unacceptable if not produced and stored in plants which meet the requirements specified in 3.7.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.1.1 Inspection of milk or milk products.

4.3.1.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the US Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest office of the Animal and Plant Health Inspection Service, Veterinary Services US Department of Agriculture. (Location of this office may be ascertained by consulting the publication entitled Directory of Meat and Poultry Inspection Program Establishments, Circuits and Officials). Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.3.1.1.2 Milk and milk product component sources. Examination shall be made to determine that all milk and milk products have been manufactured, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement or from plants having a pasteurization plant compliance rating of 90 or more as certified by a State milk sanitation rating officer and listed in the document entitled: Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers published quarterly by the U.S. Public Health Service, or listed by the U.S. Department of Agriculture in the document entitled: Dairy Plants Surveyed and Approved for USDA Grading Service and approved for the finished product made therefrom, as applicable.

4.3.1.1.3 Examination of raw whole milk. Examination and testing of raw whole milk shall be made to determine compliance with the requirements of table I, item 1, as concerns quality requirements and condition (appearance and odor). Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirements, reflected by actual examination and tests or by examination of records, shall be cause for rejection of the involved lot of finished product.

4.3.1.1.4 Examination of skimmed milk, concentrated skimmed milk, milk products used in frozen desserts, concentrated whole milk and cream (fresh or frozen) and sweet cream buttermilk. Examination and testing of these products shall be made to determine compliance with the requirements of table I, items 2, 3 and 4, as concerns source, identity, processing history, holding temperature and age at time of use. Such compliance shall be ascertained by actual examination or by examination of plant records or Standards of Identity or US Standards for Grades, or other valid documents which will reflect also the organoleptic (appearance and odor), and other referenced quality criteria of the original raw whole milk used just prior to the manufacture of these products. Failure to comply with one or more of the above referenced requirements shall be cause for rejection of the finished product made therefrom.

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4.3.1.1.5 Examination of anhydrous milk fat, butteroil, plastic cream and churned fat. These ingredients shall be examined in accordance with the applicable specification requirements for product characteristics referenced in table I, item 5. The sample for examination shall be approximately 6 ounces (170 g) of product derived from each of five primary containers or all containers if less than five form a lot. Failure of these components to comply with one or more requirements shall be cause for rejection of the involved quantity of finished product made therefrom.

4.3.1.1.6 Examination of sweetening ingredients, flavoring material, vegetable fat, colloidal stabilizers, stabilizing salts, emulsifying agents, vitamin A palmitate coloring material, yogurt culture and packaging gas. Conformance of these non-dairy ingredients to the requirements of table I, items 8, 9, 10, 11, 12, 13, 14, 16, 17, and 18 as concerns identity (kind, types, styles) shall be determined by examination of pertinent labels, invoices, US Standards for Grade or other valid documents. In addition, each of the ingredients except packaging gas shall be examined organoleptically, as necessary, to determine conformance to the applicable condition requirements of 3.1 and, as applicable, to the product characteristic requirements of MIL-F-35081 except that the sampling plans, acceptable quality levels (AQLs) and applicable Wiley Melting point shall not apply. The sample for examination shall be approximately 4 ounces (avoirdupois or fluid, as applicable) of ingredient (except dry colors for which the sample shall be 1 to 2 ounces) taken from each of five containers or all containers if the lot contains less than five containers. Nonconformance to one or more identity or condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved end item.

4.3.1.1.7 Examination of butter and dry milk products. A USDA grading certificate shall be supplied to indicate compliance with table I, items 6 and 7.

4.3.1.2 Testing of components. Testing of components described in this specification shall be as shown in table V. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE V. Testing of components (see 4.3.1)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt. para.
Cans and lids 1/	1 can and 1 lid	Cans	S-1	Tinplate	Nearest 0.01 lb/ (0.1 g/m^2) base box	4.4.3 and 5.1.1
Vegetable fat	2/	Primary containers	2/	Wiley melting point	Nearest 1°F (1°C)	3.1 4.4.1

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TABLE V. Testing of components (see 4.3.1) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt. para.
Coconut fat	2/	Primary containers	2/	Moisture and volatile matter	Nearest 0.01 percent	3.1 4.4.1
				Wiley melting point	Nearest 1°F (1°C)	
				Color	Pass or fail	
				Stability	Pass or fail 3/	
				Free fatty acids (as oleic acid)	Nearest 0.01 percent	
				Peroxide value	Nearest 0.1 meq/ kg of fat	
				Iodine	Nearest gram	
Pasteurized skim milk, sweet cream buttermilk, cream, concentrated fluid dairy products	Primary container	Primary containers	2/	Standard plate count	As specified in Standard Methods for the Examination of Dairy Products	3.1 4.4.2

1/ Exterior can coating, when required, and can label shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tinplate testing.

2/ The sample for examination shall be an approximate 1 pound composite of fat derived from five primary containers or all containers if less than five form a lot.

3/ If failure, report to nearest hour.

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4.3.2 In-process examination. Examination shall be performed during processing to determine compliance with the applicable requirements of 3.1 and 3.2, as concerns temperature, time, and time intervals, procedures, additive of vitamin A palmitate, and equipment. Records of all applicable control data shall be maintained. Noncompliance with one or more of the above referenced requirements reflected by actual examination or by records, shall be cause for rejection of the involved lot of finished product.

4.3.3 Examination of end item. Classification of defects found during examination shall be in accordance with tables VII through XII. The examination criteria for the above tables shall be as shown in table VI. AQL's shall be expressed as percent defective for table VII and in terms of defects per hundred units for other tables.

TABLE VI. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLS		
				Major A	Major B	Minor
VII	S-3	1 primary container	Primary containers	-	-	2.5
VIII	S-2	Contents of 1 primary container	Primary container	1/	1.5	-
IX	S-2	One quart of reconstituted product	Quarts	-	1.5	-
X	S-2	One quart of reconstituted product	Quarts	1/	1.5	-
XI	S-2	One quart of reconstituted product	Quarts	-	1.5	-
XII	S-2	One quart of reconstituted product	Quarts	1/	1.5	-

1/ Presence of any major A defect shall be cause for rejection of the involved lot of finished product.

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TABLE VII. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight in excess of 3/8 oz (11 g) under specified net weight. 2/

1/ Lot shall be rejected if the average net weight of the samples is less than the specified net weight.

2/ Report results to the nearest 1/8 ounce (4 g).

TABLE VIII. Examination of product characteristics, types I, II, III and IV (after reconstitution to a soft fluid serve mix, see 5.4.1) 1/

Category		Defect
Major A	Major B	
101		Presence of foreign material (e.g. dirt, insects, insect parts, wood, glass, paper, etc.).
102		Presence of ropy, curdled, whey off, watery, foamy, doughy, etc.
	151	Not type and style specified.
	152	Presence of coarseness, lumps or graininess.
	153	Texture not creamy.
	154	Body of liquid mix not viscous, not smooth or not readily flowable (see 3.3, including item 5, table III).
	155	Flavor not characteristic of the vanilla or chocolate flavor types used in commercial operations. (A mildly cooked flavor shall not be counted as a defect.) (Style A and B)
	156	The pleasing nutty-like sweet vanilla taste masks the basic acid character or vice versa (type IV, style A).
	157	Color not typical of the flavor specified (style A, B and C).

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TABLE VIII. Examination of product characteristics, types I, II, IV and III (after reconstitution to a soft fluid serve mix, see 5.4.1) 1/ (cont'd)

Category		Defect
Major A	Major B	
		Applicable to type III product prior to reconstitution:
103		Presence of objectionable flavor or odor, e.g. oxidized, stale, rancid, tallowy, cooked, scorched or fishy.
104		Presence of foreign material, e.g. dust, insects, insect parts, rodent hair or filth, wood, glass, paper, etc.
105		Not readily reconstituted (see item 4, table III and 5.4.1).
	158	Not a white or light cream color (style A).
	159	Not a light to medium shade chocolate brown color (style B).
	160	Not a fresh, sweet, pleasant flavor or aroma characteristic of the applicable style.
	161	Presence of lumps that do not fall apart under light pressure.

1/ Presence of any major A defect shall be cause for rejection of the involved lot of finished product.

TABLE IX. Examination of types I and III (style A and B) after reconstitution of soft serve ice milk.

Category	Defect
Major B	
151	Overrun not as specified (see 3.3.1.1).
152	Color not a white to light cream color (style A only).
153	Surface does not possess a dry, non-glistening surface appearance.
154	Not a stiff and firm body.
155	Product does not possess a good peaking (coiling up) characteristic.
156	Not resistable to melting.

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TABLE IX. Examination of types I and III (style A and B) after reconstitution of soft serve ice milk. (cont'd)

Category	Defect
Major B	
157	Presence of palpable ice crystals.
158	Not a smooth texture.
159	Products tends to churn in freezer cylinder.
160	Not a clean, sweet flavor characteristic to that of vanilla (style A).

TABLE X. Additional examination of types I, and III (after reconstitution, see 5.4.1) style B soft serve ice milk (see 3.3.1.2).

Category		Defect
Major A	**Major B**	
101		Product possesses a harsh, objectionable or bitter flavor.
	151	Color not a light to medium chocolate brown.
	152	Flavor not a characteristic of a clean and distinctive chocolate flavor (see 3.3.1.2, item 1).
	153	Presence of flecking (see 3.3.1.2, item 2).

TABLE XI. Additional examination of product characteristics (types II and III - see 5.4.1).

Category	Defect
Major B	
151	Overrun not as specified.
152	Shake not smooth, foamy or thick.
153	Color not white to light cream color (style A only).
154	Color not light to medium chocolate brown (style B).
155	Not a clean sweet flavor characteristic to that of vanilla (style A).

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TABLE XI. Additional examination of product characteristics (types II and III - see 5.4.1). (cont'd)

Category	Defect
Major B	
156	Flavor not characteristic of a clean and distinctive chocolate flavor (see 3.3.1.2).
157	Presence of flecking (see 3.3.1.2).
158	Formulation not flexible (see 3.3.1.3).
159	Evidence of separation, gumminess, stickiness, graininess or butteriness.
160	Quality not refreshing.
161	Product leaves a thick slimy coating in the mouth. Applicable to type III.
162	Texture not as specified in accordance with the ratio dilution factors of 3.3.1.3, item 5.

TABLE XII. Examination of soft serve yogurt (type IV)

Category		Defect
Major A	Major B	
101		Off-flavors such as, but not limited to, bitter, yeasty or diacetyl.
	151	Body is not smooth, creamy, or light to moderate viscosity.
	152	Does not have a clean, distinct lactic acid taste and characteristic vanilla flavor (style A).
	153	Does not have a clean, distinct lactic acid taste and a characteristic flavor as specified (style C).
	154	Not sufficient flavor intensity (style C).

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4.3.4 Packaging inspection.

4.3.4.1 External examination of cans and can leakage. External examination of cans and can leakage shall be performed in accordance with the applicable examination criteria of PPP-C-29.

4.3.4.2 Examination of exterior can coating (when applicable). Examination of exterior can coating shall be performed in accordance with the applicable examination criteria of TT-C-495.

4.3.4.3 Examination of can labeling. Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.3.4.4 Examination of shipping container. Examination of the filled, closed and reinforced shipping container shall be made in accordance with the applicable examination criteria in the appendix to PPP-B-636. In addition the following defects shall be included in the examination: Major: Nomenclature, NSN, contract number and dated pack missing, incorrect or illegible; Minor: (1) Other marking missing, incorrect or illegible. (2) Arrangement not as specified. (3) Sleeve missing. For level B pack, when applicable, shipping containers reinforced with other than adhesive, filament reinforced tape shall not be acceptable. When packing for level C is required to be in accordance with 5.2.3, examination shall be in accordance with the aforementioned specification except that only the defects for marking and arrangement shall apply.

4.3.4.5 Examination of palletized unit loads. When specified (see 5.3) palletized unit loads shall be examined in accordance with the criteria of MIL-L-35078.

4.3.4.6 Examination of age of product at time of shipment. Examination shall be made at time of shipment to determine compliance with 3.4. This determination shall be ascertained by examination of records, codes, marked date of pack or other valid means. Nonconformance to the above referenced requirements or age marking missing, incorrect or illegible shall be cause for rejection of the lot.

4.3.4.7 Examination of temperature at time of delivery (types I, II and IV). Examination shall be made at time of delivery to determine compliance with the temperature of 3.5. Lot size shall be expressed in containers. The sample unit shall be one container. The AQL expressed as percent defective shall be 2.5. Any container not complying with the requirement shall be considered as defective.

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4.3.5 Testing of end item. The finished product shall be tested, as applicable, for oxygen in the headspace, fat, moisture, pH (type IV), scorched particles (type IV, style A only), standard plate count, coliform count, yeast and mold (type IV), and for the presence of salmonellae in accordance with the requirements of 3.3. *Test procedures shall be in accordance with 4.5. Test requirements for fat, oxygen, and moisture and microbiological content shall be applicable to the sample unit. The sample unit for testing shall be one filled and sealed primary container. Lot size shall be expressed in terms of primary containers and the sample size shall be the number of sample units indicated by inspection level S-2. The AQL, expressed as percent defective for each characteristic except for salmonellae determination, shall be 2.5. A nonconformance to a test requirement shall be considered a defect. The sample unit for testing for scorched particles shall be approximately a 1-pound composite derived from the samples used for unit testing. Results shall be reported to the nearest milligram for scorched particles, to the nearest 0.1 percent for fat and headspace oxygen, and as positive or negative for salmonellae. All microbiological counts shall be reported in accordance with Standard Methods for the Examination of Dairy Products. The finding of any test as positive for salmonellae shall be cause for rejection of the involved lot of finished product. In addition, nonconformance to footnotes 1/, 2/, 3/, 4/ and 5/ of table IV shall be cause for rejection of the lot.

* Acceptance criterea for milk solids not fat shall be determined by review of plant records.

4.4 Test methods.

4.4.1 Chemical analyses. Chemical analyses shall be made in accordance with the methods and publications listed below:

Test	Source	Method
Ingredient:		
Coconut fat and vegetable fat, as applicable:		
Moisture and volatile matter	1/	Ca 2b-38
Melting point (Wiley method)	1/	Cc 2-38
Free fatty acid	1/	Ca 5a-40
Peroxide value	1/	Cd 8-53
Iodine value	1/	Cd 1-25
Color	1/	Cc 13b-45
Fat stability (AOM)	1/	Cd 12-57

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Test	Source	Method
Ingredient:		
Finished product:		
Fat (type I, II and IV product)	2/	Roese-Gottlieb (Official)
Fat (type III product)	2/	Fat in Dried Milk
Moisture	3/	Toluol Distillation
Scorched particles	4/	Scorched Particles (25 gram sample)
Oxygen in headspace	3/	
pH	2/ 5/	Hydrogen Ion concentration

1/ Official and Tentative Methods of the American Oil Chemists' Society, Section C, Commercial Fats and Oils.

2/ A.O.A.C. for types I, II and IV; - Chapter: Dairy Products; Section: Ice Cream and Frozen Desserts for type III; Chapter: Dairy Products; Section: Dried Milk, Nonfat Dry Milk and Malted Milks.

3/ Methods of Laboratory Analysis, Dairy Division, USDA, Da Inst. Series 918-39.

4/ American Dry Milk Institute Bulletin 916, Standards for Grades of Dry Milks, Including Methods of Analysis.

5/ AOAC methods chapter, Beverages, Section; Beer

4.4.2 Microbiological analyses. Microbiological analyses shall be performed in accordance with the applicable methods for specific dairy product components described in Standard Methods for the Examination of Dairy Products and for the finished product noted as follows:

Test	Chapter	Method
Standard plate count (types I and II)	Microbiological Methods for Concentrated Milk and Dry Milk	Condensed milk
Standard plate count (type III)	Loc. cit.	Dry milk
Coliform count (types I and II)	Loc. cit.	Test for coliform group in condensed milk
Coliform count (type III)	Loc. cit.	Test for coliform group in dry milk

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Test	Chapter	Method
Yeast and mold count (type IV)	Microbiological Methods for Cheese and Cultured Products	Test for Yeast and Mold Count
Coliform count (type IV)	Loc. Cit.	Test for Coliforms
Salmonellae test	1/	

1/ U.S. Department of Health, Education and Welfare, Bacteriological Analytical Manual, Dried Milk Section.

4.4.3 Tin coating weight. Tin coating weight shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation-packaging. Type III product shall be packaged in accordance with level A or level C, as specified (see 6.2). Type I, II and IV product shall be packaged in accordance with level C.

5.1.1 Level A. A net weight of 5 pounds (2.27 kg) of type III product shall be filled into a size 603 by 812, open-top style, round, metal can with soldered or welded side seam and compound-lined, double-seamed ends. A minus 3/8-ounce (11 g) tolerance will be allowed in any one container provided the average net weight of the cans inspected in accordance with table VII is not less than 5 pounds (2.27 kg). The can shall be made throughout from commercial 0.25-pound (5.6 g/m^2) per base box electrolytic tin plate (see 4.4.3), and shall be coated overall outside with a coating conforming to type I or, when specified (see 6.2), type III of TT-C-495. The product shall be gas-packaged by removing the air from the filled can and replacing it with nitrogen. The filled and sealed can shall not leak when tested in accordance with 4.3.4.1.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.2.1 or 5.1.2.2, as applicable.

5.1.2.1 Type I, II or IV. Type I, II or IV product shall be packaged in a manner to protect it from damage or deterioration during shipment from the supply source to the first consuming activity. The supplier may use his standard practice if it meets this requirement.

5.1.2.2 Type III. Type III product shall be packaged in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable.

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5.2 Packing. Six cans of type III product, arranged 3 in length and 2 in width, shall be packed on end in a snug-fitting shipping container in accordance with level A, B, or C, as specified (see 6.2). Type I, II or IV product shall be packed in accordance with 5.2.3.

5.2.1 Level A packing. The shipping container shall be a fiberboard box constructed, closed and reinforced in accordance with style RSC, V2s, of PPP-B-636. Alternatively, the shipping container shall be a wax impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax impregnated box shall be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. The wax impregnated boxes shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636. The reinforcing bands shall be uncolored or of a color which contrasts with black.

5.2.2 Level B packing. The shipping container shall be a fiberboard box, constructed, closed and reinforced in accordance with style RSC, V3c, V3s or V4s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax-impregnated box shall be with hot-melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. The wax-impregnated container shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636. The reinforcing bands shall be uncolored or of a color which will contract with black.

5.2.3 Level C packing. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.2), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Cans. Cans of the product shall be labeled in accordance with MIL-L-1497. Labeling shall include directions for use as follows:

FOR TYPE III, STYLE A

Reconstitution for producing a soft serve fluid mix shall be in the proportion of 5 pounds (2.27 kg) of dehydrated product to 5 quarts (4.73 L) of 40° to 60°F (4° to 16°C) potable water.

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FOR TYPE III, STYLE B

Reconstitution for producing a soft serve fluid mix shall be in the proportion of 5 pounds (2.27 kg) of dehydrated product to 4 quarts (3.79 L) of 40° to 60°F (4° to 16°C) potable water.

FREEZING INSTRUCTIONS FOR SOFT SERVE ICE MILKS (TYPES I AND III)

Freeze in soft serve type freezer to a drawing temperature of 18° to 22°F (-8° to -6°C) or when the product being frozen reaches a good, stiff peaking condition upon filling into container.

FREEZING INSTRUCTIONS FOR DIRECT DRAW MILK SHAKES

TYPES II AND III

Reconstitute dehydrated product as for soft service ice milk and freeze in same type freezer to a drawing temperature of 27° to 30°F (-3° to -1°C) or to a slushy, soft consistency, depending on preference. Reconstituted mix may be diluted as follows to attain desired shake characteristics:

Texture Condition Desired	Ratio by volume of reconstituted mix to water
Smooth and free from palpable ice crystals	No dilution
Slightly coarse	3 parts mix to 1 part water
Medium coarse	3 parts mix to 2 parts water
Coarse and icy with palpable ice crystals	1 part mix to 1 part water

5.4.1.1 Commercial labeling. When specified, (see 6.2) commerical labeling is acceptable provided the finished product complies with all other specification requirements.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the shipping container shall be marked on top in letters 7/16 inch to 1 inch (11 to 25 mm) in height) with the following precautionary marking:

DO NOT STORE ABOVE 70°F (21°C) (for type III product)
or
KEEP REFRIGERATED (32°F to 40°F (0° to 4°C))
(for type I, II and IV product)
or

For type IV product that is frozen include the following marking:

To thaw, place containers, with space intervals between them, at room temperature (70° to 75°F) (21° to 24°C) for 24 hours, or under refrigeration (35° to 40°F) (2° to 4°C) for 48 to 72 hours. Use only when completely thawed. When not in use, refrigerate at 32° to 40°F (0° to 4°C).

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5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Technical information. Types I, II and IV products covered in this specification are classified as perishable, and type III and products as nonperishable; however, storage of type III in a cool location will aid in retention of flavor quality. Fresh liquid mix meeting the requirements of this specification and using good commercial practice, has a shelf life of 5 days. Type IV frozen mix has an estimated shelf life of 6 months when stored at uniform temperature of -15° to -20°F (-26° to -29°C) and when the package is kept sealed. Type III product, when stored at 50° to 70°F (10° to 21°C), has an approximate shelf life of 1 year. Class 2, imitation labeled products are not intended for use in Continental United States. It should be noted that no specific formulation for the products described in this specification is made mandatory because of a wide range in personal preference and the difficulty to define an ideal product. Several factors affect product characteristics, but basically high total solids levels and low drawing temperatures have a pronounced effect upon stiffness and dryness. A high sugar content will require a lower drawing temperature to provide a body and dryness comparable to a mix containing less sweetening but frozen to a higher drawing temperature. The degree of thickness and smoothness in a direct draw milk shake varies considerably with formulation, drawing temperature, overrun and the degree and intensity of agitation. The lower total solids will contribute to a cold, icy shake sensation while a mix containing a high solids content will yield a thick, heavy bodied shake.

6.1.1 Type IV product. The distinguishing difference between yogurt and other cultured dairy products sold commercially in CONUS is the use of two different types of organisms in a 1:1 blend. Both organisms, Streptococcus thermophilus and Lactobacillus bulgaricus, supplement and stimulate one another (i.e. act in symbiosis) in terms of growth, acid and flavor enhancement. Culture handling requires skill and aseptic technique, along with a good culture rotation program to insure the proper balance between the organisms and to avoid airborne contamination. The flavor profile for a good yogurt product is difficult to describe; the most practical approach is to strike a balance between a desirable acid level (pH) and a pleasing and nutty-like flavor and aroma background (acetaldehyde). The difficulty becomes increasingly more apparent as the consumer's palate is masked by both the sweeteners used and the added flavors. What must be avoided is a highly sugared product and just sufficient added fruit flavor to permit the natural yogurt flavor to come through. The development of excess acidity, i.e. below pH 4.0 will tend to override the more exquisite yogurt flavor, only to be "covered up" by the use of extra sweeteners. The manufacture of soft serve yogurt is not presently an industry standardized procedure. The variables involved in the process are numerous and no attempt is made in this document to specify a definite procedure. However, the following basic guidelines are presented in an attempt toward the recognition of an acceptable manufacturing process:

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1. Dry blend all the dry ingredients (the amount of sweeteners used should not exceed 8% of the total called for the formulation).

2. Disperse into the fluid dairy products to avoid lumping.

3. Pasteurize and homogenize blend in accordance with 3.2.1.

4. Cool to 105° to 110°F (41° to 43°C).

5. Inoculate with 2% to 3% active yogurt culture starter.

6. Ripen to a pH of 4.5 to 4.6.

7. Cool to 80°F (27°C) or lower and add remainder of sweetener. (Note: Adding sugars and flavors above 80°F (27°C) will tend to induce whey separation).

8. As cooling progresses to the 40°F (4°C) level, acid development continues at a slower rate; the rate of cooling depends on temperature, size and spacing of containers, air circulation (velocity of air movement) etc.

9. The pH should now be between 4.0 and 4.4.

10. Flavor may now be added and the entire mix be subject to texturizing, using an homogenizer (no pressure) or recirculating briefly through a positive pump. An in-line texturator and screen may accomplish the same purpose.

11. The product is now ready for packaging.

12. Store at temperature below 40°F (40°C).

6.1.2 The following examples of suggested formulation will produce finished products conforming to the requirements of this specification:

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Ingredients	Type I Style A (percent)	Type I Style B (percent)	Type II Style A (percent)	Type II Style B (percent)	Type III Style A (percent)	Type III Style B (percent)	Type IV Style A (percent)	Type IV Style C (percent)
Fat	3.25	3.25	3.25	3.25	12.40	10.70	1.50	1.50
Milk solids-not-fat	12.50	12.00	11.0 3/	10.00 3/	38.50	32.00	13.00	12.00
Sucrose	11.70	14.60	9.00	12.00	36.00	39.00	10.0	10.00
Corn syrup solids	3.10	3.20	-	-	9.60	8.50	4.0	4.00
Chocolate flavoring	-	1/	-	1/	-	8.00	-	-
Vanilla flavoring	1/	1/	1/	1/	1/	1/	1/	-
Stabilizer	2/	2/	2/	2/	2/	2/	2/	2/
Emulsifier	2/	2/	-	-	2/	2/	2/	2/
Total Solids (Approx.)	32.00	37.00	24.00 3/	28.00 3/	98.0	98.0	30.0	30.00

1/ Vanilla flavoring, depending on type and strength, should be used in amounts commensurate with flavor manufacturers' recommendations or sufficiently to provide a pleasing consumer acceptable flavor intensity. The use of vanilla flavoring in chocolate products is optional. Harsh flavor tones, particularly characteristic of vanillin, are to be avoided. Chocolate flavor, depending on types used, in a range of from 1.5 to 3.0 percent is generally acceptable as providing a characterizing and pleasing chocolate flavor intensity.

2/ The amount of stabilizer-emulsifier used in the liquid mix or prior to dehydration may be in the range of from 0.3 to 0.5 percent.

3/ For a smooth shake drink, a solids-not-fat content of 13.00 percent (style A) or 11.50-12.00 percent (style B) is recommended. Corresponding total solids will run about 28.00 percent (style A) or 29.00-30.00 percent (style B), respectively. For a coarse shake, solids content may be cut to less than 11.00 percent solids-not-fat, but not generally below 8.50 percent for style A. Style B solids-not-fat may run as low as 9.00 percent.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class and style of product required (see 1.2).
(c) When vegetable fat may be used (see 3.1).
(d) When type III product may be delivered more than 60 days after date of production (see 3.4).
(e) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(f) Type of exterior can coating required (see 5.1).
(g) When packing specified in 5.2.2.1 or 5.2.3.1 is required.
(h) Type and class of unit load when specified (see 5.3).
(i) When commercial labeling is acceptable (see 5.4.1.1).

6.3 Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.33A, or DSAR 4145.7, as applicable.

6.4 Packing specified in 5.2.3 is intended for direct shipment from the supply source to the consuming activity for immediate use within CONUS.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

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Navy - NU
Air Force - 50

Preparing activity:

Army - GL

Project No. 8910-0349

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-43717C; ICE MILK, MILK SHAKE AND YOGURT MIXES

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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